



QUICK-RELEASE BEAR SPRAY



DEFINING THE PROBLEM

Bear charges:

- 45-50 km/h

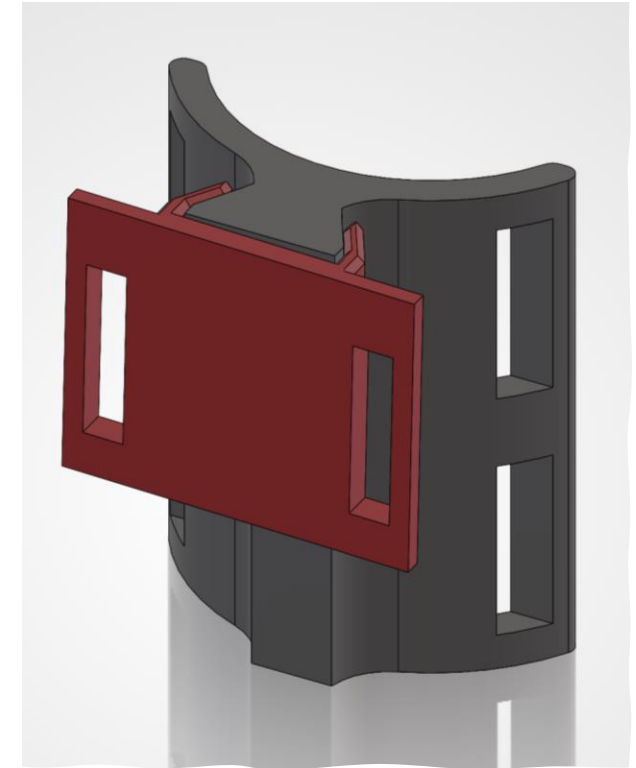
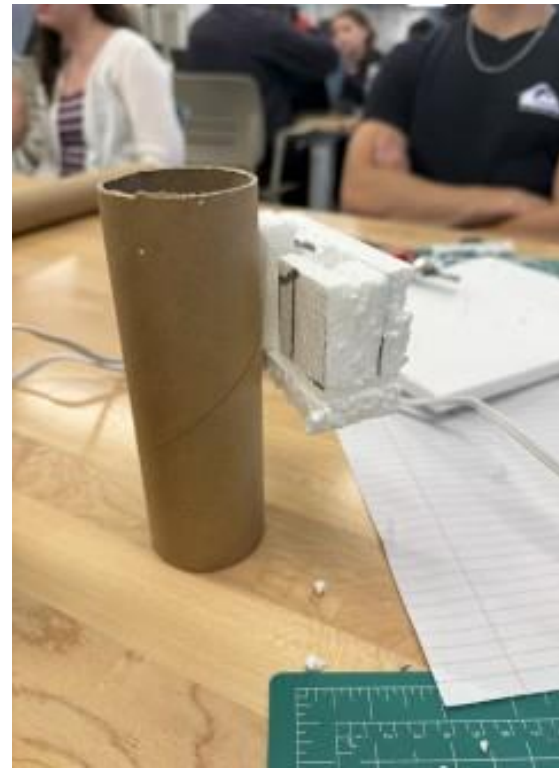
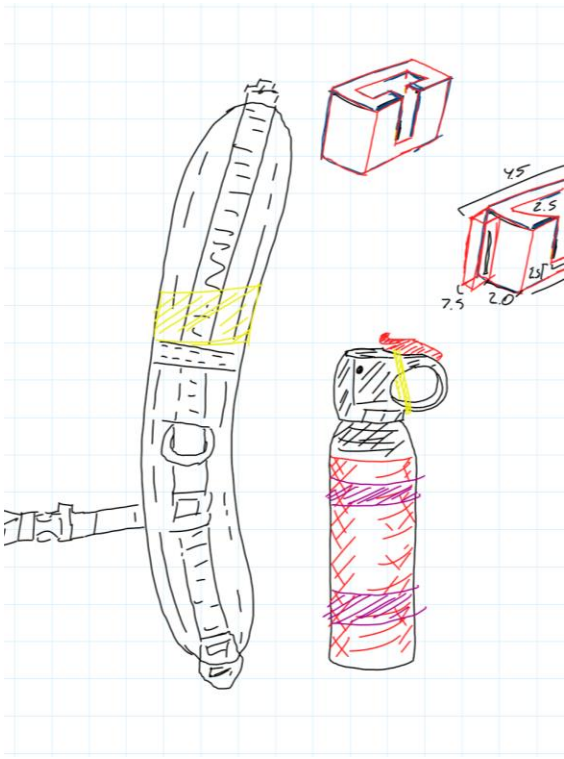
Other Design Issues:

- The bottle swings around
- Constrained by zippers
- Difficult to release
- Aren't universal to all backpacks



"A surprise bear encounter can happen in seconds"

- Parks Canada

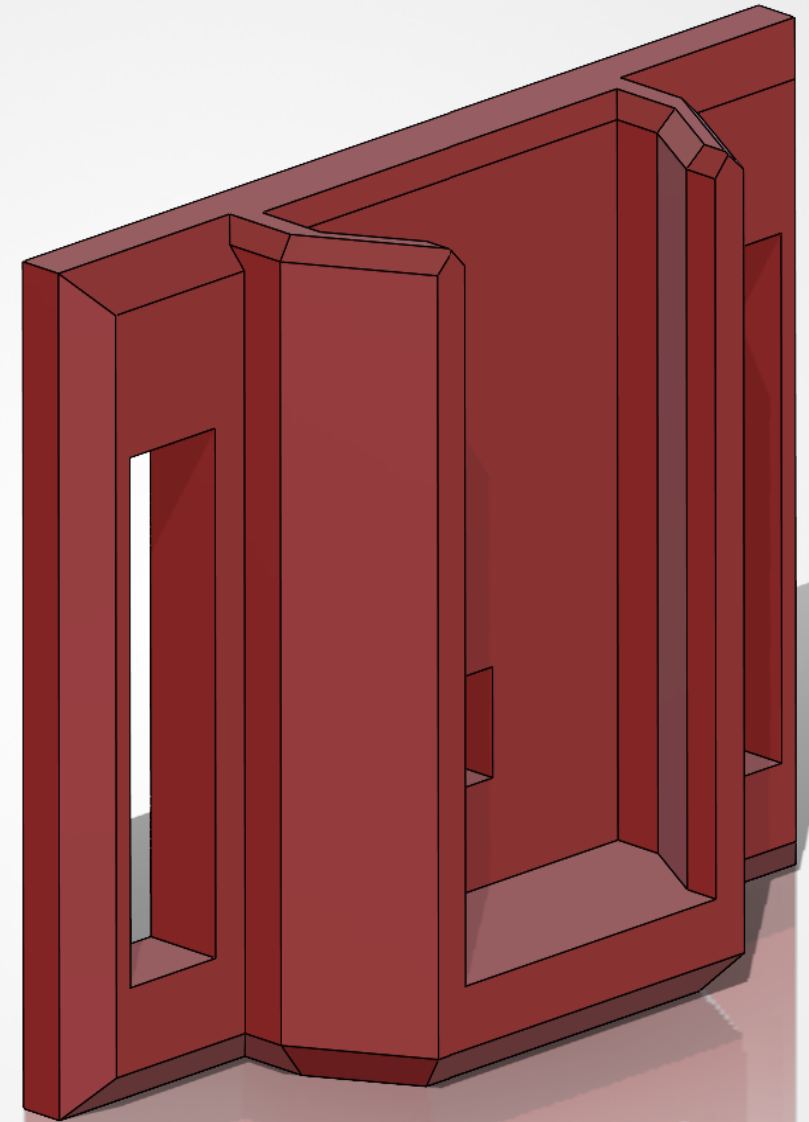
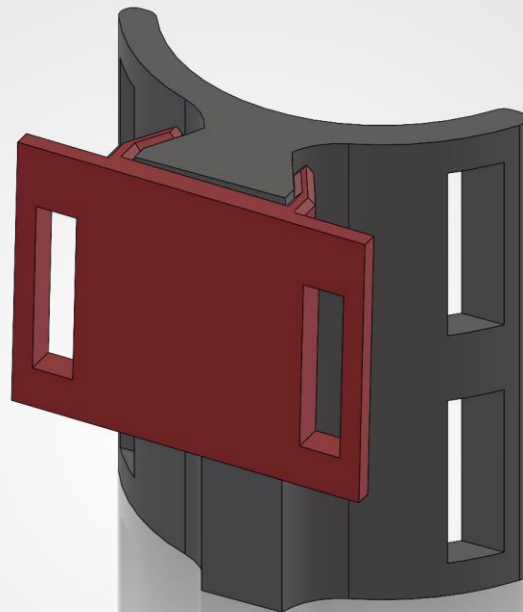
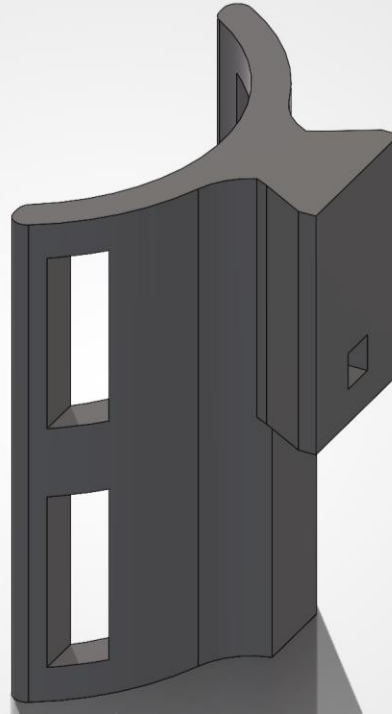


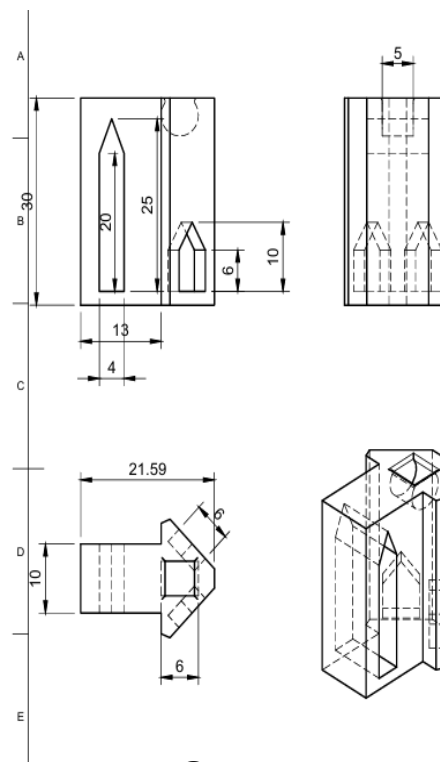
IDEATION & DESIGN PROCESS

- Market Research
- Concept Development
- Design Selection
- CAD Development

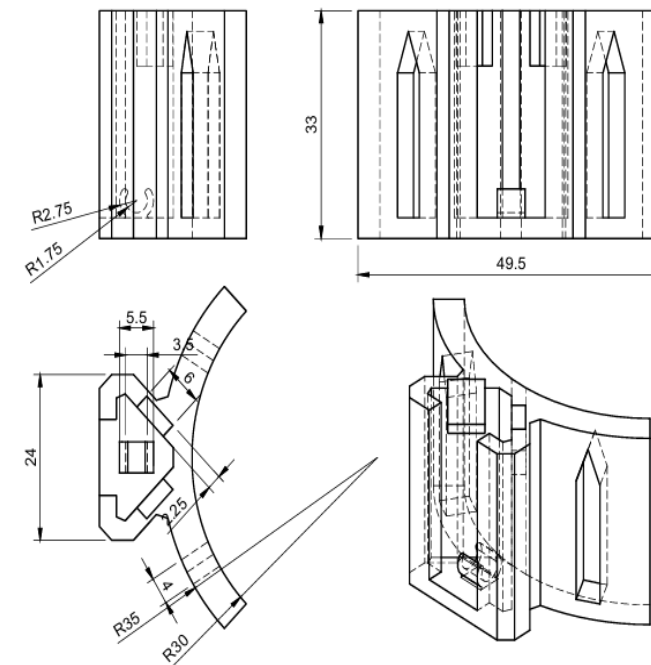
DESIGN SPECIFICATIONS

- Quick release mechanism
- Secure
- Reliable and intuitive
- Easily manufactured





Component 1



Component 2

SOLUTION

- Lightweight
- Easy to use
- Dovetail
- Clips & Magnets
- Friction

MOVING FORWARD

Future Improvements:

- New mounts for new applications:
 - Bike mount
 - Car mount
 - Belt mount

Further Industry Use Cases:

- Health Care
- Defense
- Skilled Trade

